Superconducting Devices

From wastewater treatment to MEG and buried-metal detection

One physics, two device families, three applications

Filippo Vicidomini 884022

University Ca’ Foscari of Venice

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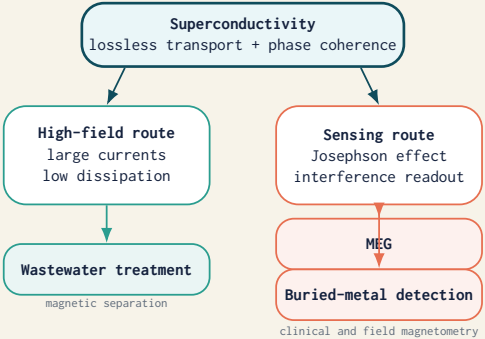
Guiding idea

- **Two routes**: high current and phase-coherent sensing
- **Two device families**: superconducting magnets and SQUIDs
- **One claim**: shared physics, different engineering targets

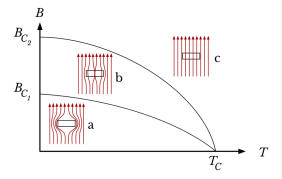


What makes a superconductor technologically useful?

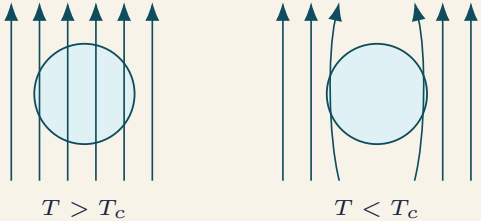
- $R = 0$: persistent currents
- **Meissner**: true superconducting phase
- **Type II**: vortices for $H_{c1} < H < H_{c2}$
- **Mixed state**: route to high fields

$$\nabla^2 \mathbf{B} = \frac{\mathbf{B}}{\lambda_L^2} \qquad \Phi_0 = \frac{h}{2e}$$

Takeaway: The same material platform supports both strong magnetic action and precise magnetic control.



Type-I screening, vortex mixed state, and field penetration above H_{c2}.



Meissner effect: below T_c the magnetic field is expelled.

Macroscopic phase and the Josephson effect

- **Order parameter**

$$\psi(\mathbf{r}) = |\psi(\mathbf{r})| e^{i\theta(\mathbf{r})}$$

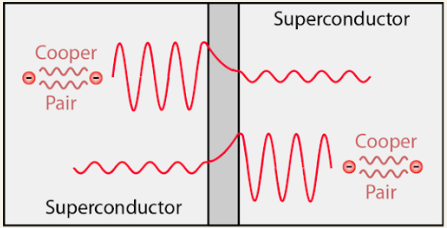
- **Phase θ is observable**
- **Josephson current**

$$I_s = I_c \sin\varphi$$

- **Voltage drives phase**

$$\frac{d\varphi}{dt} = \frac{2e}{\hbar} V$$

Phase difference $\longrightarrow$ current voltage $\longrightarrow$ phase evolution



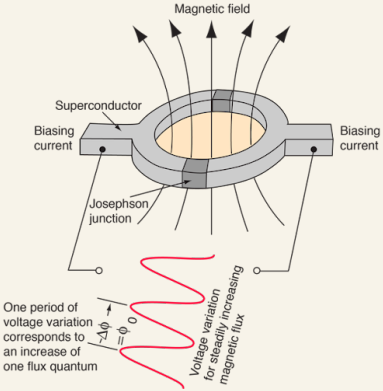
Simple SIS Josephson junction: the phase difference across the barrier controls the supercurrent.

SQUID: flux-to-voltage transducer

- **Loop + 2 JJ**
- **Flux sets interference**
- **Periodic transfer curve**

$$I_c(\Phi) \simeq 2I_0 \left| \cos\left(\pi \frac{\Phi}{\Phi_0}\right) \right|$$

$\Delta\Phi \to \Delta V$



Takeaway: Quantum interference turns tiny flux changes into readable voltage modulation.

From phase to measurement (intuition)

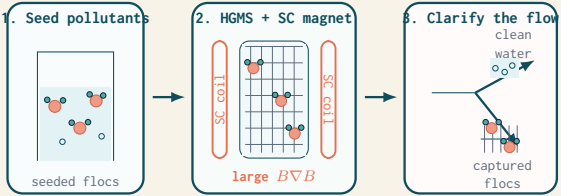
$$B \to \Phi \to \varphi \to I_s \to V$$

A SQUID reads **voltage** at the output, but that voltage comes from a chain that starts with the magnetic field.

Takeaway: The measured voltage is the end of a transduction chain, not the starting physical quantity.

Application 1: wastewater treatment by magnetic separation

- Tag pollutants with magnetic seeds
- HGMS matrix traps magnetic flocs
- Clean water exits, flocs stay
- **Need**: strong magnetic force → large $B\nabla B$



Example: up to 76% COD removal

Paper mill wastewater, Zhang et al. (2011).

Takeaway: Strong-field branch: superconductivity provides magnetic action efficiently.

Why use a superconducting magnet here?

- Resistive magnet: high field, high power cost
- SC coil: stable field, low Joule loss
- Better for continuous operation
- Main constraint: cryogenics + process integration

Message

Here superconductivity is used to generate force, not to measure.

	Resistive	SC
High field	possible	practical
Continuous duty	costly	favorable
Power dissipation	high	low in the coil
System constraint	power supply	cryogenics

Application 2: magnetoencephalography with SQUID arrays

- Brain fields down to 10^{-15} T
- Whole-head SQUID arrays
- Epilepsy + functional mapping
- Sensitivity + ms resolution

Takeaway: In this branch the limiting factor is not force, but the ability to read out extraordinarily weak fields non-invasively.

Image placeholder

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Expected name: meg-system.png

Application 3: buried-metal and landmine detection

- Excitation coil → eddy currents
- Target → weak secondary field
- SQUID advantage: low-frequency sensitivity
- Field issue: noise + motion + cryogenics

Takeaway: This is still the sensing branch, but now in a much noisier and less controlled environment than MEG.

Image placeholder

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Expected name: buried-metal-detection.png

One physics, three engineering targets

Application	Useful quantity	Why superconductivity helps	Main constraint
Wastewater treatment	High field / gradient	Magnetic separation with low coil losses	Cryogenics
MEG	fT biomagnetism	SQUID flux readout	Shielding
Buried-metal detection	Weak LF secondary field	Low-frequency sensitivity	Field robustness

Takeaway: The unity comes from superconducting physics; the diversity comes from what the device is optimized to do.

Limits and outlook

Current limits

- Cryogenic infrastructure
- Magnetic shielding and noise control
- System complexity outside the lab
- Cost and maintenance

Where progress matters

- Better HTS materials and devices
- Simpler cryocooler-based operation
- More robust low-noise electronics
- Easier deployment in clinical and field environments

Takeaway: The open challenge is no longer only physical understanding, but engineering simplification and scale-up.

Conclusion

- **Create field**: superconducting magnets
- **Read field**: SQUIDs
- **Same physics**, three applications

Take-home message:

Superconductivity becomes a technology platform because it supports both magnetic action and magnetic readout.

References I

Course material

- R. Arpaia, Lecture 4 – Superconductivity. Key Phenomena and Concepts.
- R. Arpaia, Lecture 7 – Superconductivity. London Equations (I).
- R. Arpaia, Lecture 15 – The Josephson Effect. Gateway to Superconducting Device Physics (II).
- R. Arpaia, Lecture 18 – Josephson Junction Devices. From Quantum Qubits to SQUIDs in Neuroscience.

Applications

- Zhang, Y. et al., Study on industrial wastewater treatment using superconducting magnetic separation, Cryogenics 51 (2011) 225–228.
- Li, J. et al., FeOOH and nZVI combined with superconducting high gradient magnetic separation for the remediation of high-arsenic metallurgical wastewater, Separation and Purification Technology 285 (2022) 120372.
- Hari, R. et al., IFCN-endorsed practical guidelines for clinical magnetoencephalography (MEG), Clinical Neurophysiology 129 (2018) 1720–1747.

References II

- Guo, X. et al., Metal detector based on high-T_c RF SQUID, Physica C 378–381 (2002) 1404–1407.